EVOLUTIONARY COMPUTATION

LABORATORY 02

Greedy regret heuristics

Selecting a subset and arranging it into a low-cost Hamiltonian cycle

Focus

2-regret selection

and weighted insertion scores

CORE OBJECTIVE

Minimize cycle length + selected node costs

Every solution selects ceil(n / 2) nodes and closes the route with an edge back to the first selected node.

INSTANCES

TSPA - TSPB

200 nodes each - 100 selected

EXPERIMENT

8 constructive methods

200 evaluated solutions per method

Problem and experimental design

Each node has integer coordinates and a selection cost.

The solver chooses k = ceil(n / 2) distinct nodes and orders them into a closed Hamiltonian cycle. Distances are rounded Euclidean distances computed before optimization.

OBJECTIVE FUNCTION

F(C) = cycle length(C) + sum of selected node costs

CONSTRAINTS

- select exactly ceil(n / 2) nodes
- visit each selected node once
- return to the first node
- use zero-based indices

EXPERIMENT

TSPA and TSPB

200 nodes / instance

100 selected nodes

200 starts / method

Recorded outputs

For every instance and method, the program writes min, max, average, standard deviation, runtime, best and worst routes, and SVG visualizations. Every produced route is checked internally for size, closure, range, and uniqueness.

Shared representation

Optimization code receives the precomputed distance matrix and node costs.

Coordinates are used only for distance construction and final 2D visualization.

This matches the separation required by the problem description.

Algorithms and pseudocode

The regret value measures how costly it would be to postpone a node.

2-regret on an open path

```
path <- [start]
while |path| < k:
 for each unused v:
evaluate every path insertion
best <- smallest delta(v)
second <- second smallest delta(v)
regret(v) <- second - best
choose max regret(v)
insert v at its best position
```

2-regret on a closed cycle

```
cycle <- [start, best second node]
while |cycle| < k:
 for each unused v and cycle edge:
delta <- d(a,v)+d(v,b)-d(a,b)+c(v)
keep the two best deltas
regret(v) <- second - best
choose max regret(v)
insert v at its best edge
```

Weighted score - path

```
regret(v) <- second(v) - best(v)
score(v) <- wr * regret(v)
 - wd * best(v)
choose the maximum score
default: wr = 1, wd = 1
insert at the best path position
```

Weighted score - cycle

```
regret(v) <- second(v) - best(v)
score(v) <- wr * regret(v)
 - wd * best(v)
choose the maximum score
default: wr = 1, wd = 1
insert at the best cycle edge
```

Insertion delta

For both representations, delta includes the change in cycle/path length plus the newly selected node cost. The path variants follow lab 01's open-path insertion convention; cycle variants evaluate closed-cycle edges.

Computational experiment - TSPA

Each row aggregates 200 solutions generated from all possible starting nodes.

Weighted methods use equal weights: regret_weight = 1 and delta_weight = 1.

Objective statistics - TSPA

Method	Min	Max	Average	Std. dev.
Random solution	237026	285754	261754.81	9444.81
Nearest neighbor - end	84362	90075	87476.63	1328.22
Nearest neighbor - any position	72328	77022	74347.62	981.22
Greedy cycle	72590	75281	73793.41	527.45
2-regret - path	112968	131943	121241.39	3928.02
2-regret - cycle	111748	129972	120502.06	3493.64
Weighted regret - path	72307	78176	75189.33	1612.85
Weighted regret - cycle	72011	76027	73335.27	869.43

Result summary

Best individual solution: Weighted regret - cycle with objective 72011.

Best average: Weighted regret - cycle with average objective 73335.27.

Runtime (all 200 solutions)

Method	Runtime
Random solution	2 ms
Nearest neighbor - end	88 ms
Nearest neighbor - any position	21143 ms
Greedy cycle	11978 ms
2-regret - path	21526 ms
2-regret - cycle	12376 ms
Weighted regret - path	21631 ms
Weighted regret - cycle	12443 ms

Computational experiment - TSPB

Each row aggregates 200 solutions generated from all possible starting nodes.

Weighted methods use equal weights: regret_weight = 1 and delta_weight = 1.

Objective statistics - TSPB

Method	Min	Max	Average	Std. dev.
Random solution	181550	228271	206936.33	8541.75
Nearest neighbor - end	56007	63166	59397.34	2093.12
Nearest neighbor - any position	52915	66164	59465.47	2193.52
Greedy cycle	55655	63734	59649.91	2804.92
2-regret - path	67937	79102	73139.61	2178.64
2-regret - cycle	66167	77274	72327.22	1916.47
Weighted regret - path	51301	65594	57804.59	2791.11
Weighted regret - cycle	52860	63406	57903.04	3551.12

Result summary

Best individual solution: Weighted regret - path with objective 51301.

Best average: Weighted regret - path with average objective 57804.59.

Runtime (all 200 solutions)

Method	Runtime
Random solution	3 ms
Nearest neighbor - end	90 ms
Nearest neighbor - any position	21330 ms
Greedy cycle	12329 ms
2-regret - path	21873 ms
2-regret - cycle	12814 ms
Weighted regret - path	21845 ms
Weighted regret - cycle	12763 ms

Best solutions - TSPA

Node cost is encoded by colour and size; the polyline is the closed route.

Random solution

min 237026 | max 285754 | avg 261754.81

Best route

```
8 122 67 57 124 25 78 54 129 32 192 9 179 18 178 17 37 111 73 143 113 29
152 84 3 136 44 191 30 171 170 13 142 48 199 181 85 55 108 198 36 10 65 62
19 14 169 79 49 116 41 11 98 4 31 66 63 96 164 120 186 45 52 94 81 134
159 5 80 91 46 59 163 109 33 190 174 112 22 148 141 175 135 83 131 160 6 125
146 58 47 139 119 35 28 196 68 115 20 144 8
```

Nearest neighbor - end

min 84362 | max 90075 | avg 87476.63

Best route

```
54 9 33 36 177 67 144 53 135 58 134 176 174 106 89 107 57 182 151 93 50 3
1 164 178 29 136 186 184 47 159 130 153 78 88 193 170 146 75 64 150 42 113 43
85 142 8 131 84 158 192 181 114 141 180 132 76 80 48 31 63 149 168 97 127 187
30 123 152 108 60 116 49 11 119 160 83 90 51 16 35 87 45 140 52 179 185 28
102 156 157 71 91 38 100 4 112 139 115 5 54
```

Nearest neighbor - any position

min 72328 | max 77022 | avg 74347.62

Best route

```
171 2 189 30 187 141 87 45 73 135 114 181 192 158 58 84 134 72 176 150 151 182
57 64 107 88 78 193 153 170 146 159 130 113 89 42 106 93 47 147 50 3 169 1
178 101 29 163 136 186 164 8 174 131 54 9 33 36 177 52 179 53 140 144 67 16
51 156 35 185 102 28 180 132 76 4 71 157 91 80 48 31 60 108 152 100 123 83
160 49 11 63 149 116 168 97 127 69 129 104 171
```

Greedy cycle

min 72590 | max 75281 | avg 73793.41

Best route

```
48 80 91 157 71 4 76 132 180 28 102 185 35 156 51 16 67 144 140 53 179 52
177 36 33 9 54 131 174 8 142 164 186 136 163 29 101 178 1 169 3 50 147 47
93 106 42 89 113 130 159 146 170 153 193 78 88 107 64 57 182 151 150 176 72 134
84 58 158 192 181 114 135 73 45 87 141 187 30 189 2 171 69 127 97 160 83 123
90 100 152 108 60 49 11 63 168 149 116 31 48
```

Best solutions - TSPA

Node cost is encoded by colour and size; the polyline is the closed route.

2-regret - path

min 112968 | max 131943 | avg 121241.39

Best route

```
4 76 132 68 90 83 160 12 23 30 187 141 87 102 185 67 144 179 140 73 135 58
84 99 36 33 143 62 190 112 115 175 39 154 34 109 186 136 24 163 19 29 17 101
178 59 155 21 147 37 47 117 130 159 170 75 193 122 107 103 42 106 81 128 142 194
174 176 0 138 182 111 158 192 181 118 114 25 22 98 70 96 10 183 166 2 32 92
161 120 97 49 108 60 116 119 168 188 69 104 4
```

2-regret - cycle

min 111748 | max 129972 | avg 120502.06

Best route

```
171 187 30 23 162 90 123 152 108 11 97 120 110 129 168 116 82 77 199 48 126 137
80 38 91 157 4 196 51 133 139 112 5 115 175 39 154 62 34 109 186 136 197 163
19 29 17 55 86 155 21 147 47 159 130 117 93 50 1 128 164 194 174 81 43 113
89 124 107 75 26 78 88 64 57 105 182 138 111 192 58 135 73 41 99 177 52 144
45 185 102 87 96 98 114 22 10 183 166 2 171
```

Weighted regret - path

min 72307 | max 78176 | avg 75189.33

Best route

```
49 108 152 100 123 83 160 30 187 171 2 189 141 87 45 73 135 114 181 192 158 58
84 134 72 176 174 150 106 113 89 42 151 138 182 57 64 107 88 78 193 153 75 170
146 159 130 47 93 50 3 169 1 178 29 163 136 186 164 142 8 9 54 33 131 36
177 52 179 53 140 144 67 16 51 35 185 102 28 180 132 76 4 71 157 91 38 80
48 31 60 116 149 63 168 97 127 69 129 104 49
```

Weighted regret - cycle

min 72011 | max 76027 | avg 73335.27

Best route

```
60 116 149 63 168 11 49 108 152 100 123 83 160 97 127 161 30 187 171 2 189 141
87 45 73 135 114 181 192 158 58 84 134 72 176 174 150 106 113 89 42 151 138 182
57 64 107 88 78 193 153 75 170 146 159 130 47 93 50 3 169 1 178 101 55 29
163 136 186 164 142 8 9 54 33 131 36 177 52 179 53 140 144 67 16 51 35 185
102 28 180 132 76 4 71 157 91 80 48 31 60
```

Best solutions - TSPB

Node cost is encoded by colour and size; the polyline is the closed route.

Random solution

min 181550 | max 228271 | avg 206936.33

Best route

66 173 10 119 113 100 71 55 44 114 85 87 160 68 2 199 189 80 162 24 29 179
34 133 191 150 46 109 56 93 62 174 67 108 60 22 130 19 78 156 65 26 134 40
177 48 7 49 102 148 9 106 12 50 158 21 95 186 90 123 96 181 74 198 88 129
165 154 155 161 97 195 111 69 79 175 180 122 127 104 182 39 25 115 53 132 57 185
5 140 4 32 43 91 89 54 116 136 121 45 66

Nearest neighbor - end

min 56007 | max 63166 | avg 59397.34

Best route

89 57 76 54 127 129 90 153 20 14 2 110 118 56 114 196 10 148 143 78 192 162
104 185 184 3 151 69 172 133 120 96 121 198 48 72 46 86 117 160 136 8 21 154
140 47 170 125 177 6 132 29 197 175 134 45 60 40 71 155 97 115 180 65 28 149
145 158 116 147 42 50 130 30 183 38 64 85 59 167 51 87 179 82 88 171 99 67
24 34 181 152 73 98 27 135 75 79 156 43 89

Nearest neighbor - any position

min 52915 | max 66164 | avg 59465.47

Best route

82 87 179 181 24 34 152 167 73 51 85 59 64 168 189 183 30 176 21 38 47 140
154 97 141 40 60 157 45 134 175 71 155 115 180 6 29 132 36 197 170 26 125 177
158 65 145 28 149 13 146 37 116 147 42 50 191 130 121 198 96 48 43 156 120 133
136 172 69 160 8 117 122 151 173 184 3 72 46 86 89 57 185 76 104 162 192 54
80 127 129 90 52 182 153 74 20 14 2 110 82

Greedy cycle

min 55655 | max 63734 | avg 59649.91

Best route

174 36 197 6 180 115 170 97 140 47 155 71 175 134 45 157 60 141 40 33 64 21
189 168 69 160 172 136 133 120 8 117 72 46 86 184 3 173 122 151 98 111 27 135
76 185 162 192 80 52 182 20 74 2 93 110 118 14 153 90 129 127 54 104 57 89
156 43 48 96 198 121 63 130 191 183 30 176 38 154 26 125 177 138 158 145 149 147
42 116 37 146 13 28 65 178 103 29 132 1 174

Best solutions - TSPB

Node cost is encoded by colour and size; the polyline is the closed route.

2-regret - path

min 67937 | max 79102 | avg 73139.61

Best route

150 52 129 127 106 162 54 78 10 114 193 61 105 169 165 199 187 70 190 194 77 57
83 44 89 32 156 166 92 120 63 96 198 121 195 50 42 116 123 15 158 149 13 28
145 65 178 103 132 1 108 36 91 62 175 134 45 157 60 141 155 113 5 180 115 170
125 26 161 154 140 47 38 21 183 168 172 69 124 4 33 107 85 51 82 88 126 181
25 98 151 173 184 137 86 75 135 76 144 104 150

2-regret - cycle

min 66167 | max 77274 | avg 72327.22

Best route

41 196 114 148 143 131 118 68 110 23 93 9 188 14 20 74 17 171 39 99 163 88
82 126 152 167 51 85 107 49 33 47 140 154 161 26 125 170 97 155 141 60 157 45
134 175 62 91 36 197 5 6 102 103 178 65 145 28 149 13 37 116 42 195 130 30
21 168 69 172 133 120 63 198 58 48 156 166 72 46 109 184 173 151 111 27 135 0
79 185 144 162 192 106 84 150 80 52 90 78 41

Weighted regret - path

min 51301 | max 65594 | avg 57804.59

Best route

121 198 96 48 89 156 120 133 136 172 160 69 8 117 72 46 86 184 3 122 151 98
27 135 76 185 57 104 162 54 192 127 90 129 80 52 182 153 14 2 20 74 17 171
39 99 150 84 66 67 24 34 152 181 88 179 87 82 51 167 73 59 85 45 134 175
157 60 40 141 71 155 115 170 97 47 21 38 140 154 26 125 177 19 6 197 29 132
103 178 65 145 28 149 13 146 37 116 147 42 121

Weighted regret - cycle

min 52860 | max 63406 | avg 57903.04

Best route

5 197 29 132 6 19 177 125 26 154 140 47 38 21 30 183 191 130 63 121 198 96
48 43 156 32 89 57 79 185 104 54 196 114 10 90 14 2 74 20 153 129 127 80
192 162 76 135 27 11 66 67 24 34 152 181 126 88 179 87 82 167 51 85 59 73
98 151 122 173 3 184 86 46 72 117 8 120 133 136 172 160 69 124 4 64 33 40
141 60 157 45 134 175 71 155 97 170 115 180 5

Best solution node lists - TSPA

Routes are zero-based and include the repeated first node that closes the cycle.

Random solution

8 122 67 57 124 25 78 54 129 32 192 9 179 18 178 17 37 111 73 143 113 29 152 84 3 136 44
191 30 171 170 13 142 48 199 181 85 55 108 198 36 10 65 62 19 14 169 79 49 116 41 11 98 4
31 66 63 96 164 120 186 45 52 94 81 134 159 5 80 91 46 59 163 109 33 190 174 112 22 148 141
175 135 83 131 160 6 125 146 58 47 139 119 35 28 196 68 115 20 144 8

Nearest neighbor - end

54 9 33 36 177 67 144 53 135 58 134 176 174 106 89 107 57 182 151 93 50 3 1 164 178 29 136
186 184 47 159 130 153 78 88 193 170 146 75 64 150 42 113 43 85 142 8 131 84 158 192 181 114 141
180 132 76 80 48 31 63 149 168 97 127 187 30 123 152 108 60 116 49 11 119 160 83 90 51 16 35
87 45 140 52 179 185 28 102 156 157 71 91 38 100 4 112 139 115 5 54

Nearest neighbor - any position

171 2 189 30 187 141 87 45 73 135 114 181 192 158 58 84 134 72 176 150 151 182 57 64 107 88 78
193 153 170 146 159 130 113 89 42 106 93 47 147 50 3 169 1 178 101 29 163 136 186 164 8 174 131
54 9 33 36 177 52 179 53 140 144 67 16 51 156 35 185 102 28 180 132 76 4 71 157 91 80 48
31 60 108 152 100 123 83 160 49 11 63 149 116 168 97 127 69 129 104 171

Greedy cycle

48 80 91 157 71 4 76 132 180 28 102 185 35 156 51 16 67 144 140 53 179 52 177 36 33 9 54
131 174 8 142 164 186 136 163 29 101 178 1 169 3 50 147 47 93 106 42 89 113 130 159 146 170 153
193 78 88 107 64 57 182 151 150 176 72 134 84 58 158 192 181 114 135 73 45 87 141 187 30 189 2
171 69 127 97 160 83 123 90 100 152 108 60 49 11 63 168 149 116 31 48

2-regret - path

4 76 132 68 90 83 160 12 23 30 187 141 87 102 185 67 144 179 140 73 135 58 84 99 36 33 143
62 190 112 115 175 39 154 34 109 186 136 24 163 19 29 17 101 178 59 155 21 147 37 47 117 130 159
170 75 193 122 107 103 42 106 81 128 142 194 174 176 0 138 182 111 158 192 181 118 114 25 22 98 70
96 10 183 166 2 32 92 161 120 97 49 108 60 116 119 168 188 69 104 4

2-regret - cycle

171 187 30 23 162 90 123 152 108 11 97 120 110 129 168 116 82 77 199 48 126 137 80 38 91 157 4
196 51 133 139 112 5 115 175 39 154 62 34 109 186 136 197 163 19 29 17 55 86 155 21 147 47 159
130 117 93 50 1 128 164 194 174 81 43 113 89 124 107 75 26 78 88 64 57 105 182 138 111 192 58
135 73 41 99 177 52 144 45 185 102 87 96 98 114 22 10 183 166 2 171

Weighted regret - path

49 108 152 100 123 83 160 30 187 171 2 189 141 87 45 73 135 114 181 192 158 58 84 134 72 176 174
150 106 113 89 42 151 138 182 57 64 107 88 78 193 153 75 170 146 159 130 47 93 50 3 169 1 178
29 163 136 186 164 142 8 9 54 33 131 36 177 52 179 53 140 144 67 16 51 35 185 102 28 180 132
76 4 71 157 91 38 80 48 31 60 116 149 63 168 97 127 69 129 104 49

Weighted regret - cycle

60 116 149 63 168 11 49 108 152 100 123 83 160 97 127 161 30 187 171 2 189 141 87 45 73 135 114
181 192 158 58 84 134 72 176 174 150 106 113 89 42 151 138 182 57 64 107 88 78 193 153 75 170 146
159 130 47 93 50 3 169 1 178 101 55 29 163 136 186 164 142 8 9 54 33 131 36 177 52 179 53
140 144 67 16 51 35 185 102 28 180 132 76 4 71 157 91 80 48 31 60

Best solution node lists - TSPB

Routes are zero-based and include the repeated first node that closes the cycle.

Random solution

66 173 10 119 113 100 71 55 44 114 85 87 160 68 2 199 189 80 162 24 29 179 34 133 191 150 46
109 56 93 62 174 67 108 60 22 130 19 78 156 65 26 134 40 177 48 7 49 102 148 9 106 12 50
158 21 95 186 90 123 96 181 74 198 88 129 165 154 155 161 97 195 111 69 79 175 180 122 127 104 182
39 25 115 53 132 57 185 5 140 4 32 43 91 89 54 116 136 121 45 66

Nearest neighbor - end

89 57 76 54 127 129 90 153 20 14 2 110 118 56 114 196 10 148 143 78 192 162 104 185 184 3 151
69 172 133 120 96 121 198 48 72 46 86 117 160 136 8 21 154 140 47 170 125 177 6 132 29 197 175
134 45 60 40 71 155 97 115 180 65 28 149 145 158 116 147 42 50 130 30 183 38 64 85 59 167 51
87 179 82 88 171 99 67 24 34 181 152 73 98 27 135 75 79 156 43 89

Nearest neighbor - any position

82 87 179 181 24 34 152 167 73 51 85 59 64 168 189 183 30 176 21 38 47 140 154 97 141 40 60
157 45 134 175 71 155 115 180 6 29 132 36 197 170 26 125 177 158 65 145 28 149 13 146 37 116 147
42 50 191 130 121 198 96 48 43 156 120 133 136 172 69 160 8 117 122 151 173 184 3 72 46 86 89
57 185 76 104 162 192 54 80 127 129 90 52 182 153 74 20 14 2 110 82

Greedy cycle

174 36 197 6 180 115 170 97 140 47 155 71 175 134 45 157 60 141 40 33 64 21 189 168 69 160 172
136 133 120 8 117 72 46 86 184 3 173 122 151 98 111 27 135 76 185 162 192 80 52 182 20 74 2
93 110 118 14 153 90 129 127 54 104 57 89 156 43 48 96 198 121 63 130 191 183 30 176 38 154 26
125 177 138 158 145 149 147 42 116 37 146 13 28 65 178 103 29 132 1 174

2-regret - path

150 52 129 127 106 162 54 78 10 114 193 61 105 169 165 199 187 70 190 194 77 57 83 44 89 32 156
166 92 120 63 96 198 121 195 50 42 116 123 15 158 149 13 28 145 65 178 103 132 1 108 36 91 62
175 134 45 157 60 141 155 113 5 180 115 170 125 26 161 154 140 47 38 21 183 168 172 69 124 4 33
107 85 51 82 88 126 181 25 98 151 173 184 137 86 75 135 76 144 104 150

2-regret - cycle

41 196 114 148 143 131 118 68 110 23 93 9 188 14 20 74 17 171 39 99 163 88 82 126 152 167 51
85 107 49 33 47 140 154 161 26 125 170 97 155 141 60 157 45 134 175 62 91 36 197 5 6 102 103
178 65 145 28 149 13 37 116 42 195 130 30 21 168 69 172 133 120 63 198 58 48 156 166 72 46 109
184 173 151 111 27 135 0 79 185 144 162 192 106 84 150 80 52 90 78 41

Weighted regret - path

121 198 96 48 89 156 120 133 136 172 160 69 8 117 72 46 86 184 3 122 151 98 27 135 76 185 57
104 162 54 192 127 90 129 80 52 182 153 14 2 20 74 17 171 39 99 150 84 66 67 24 34 152 181
88 179 87 82 51 167 73 59 85 45 134 175 157 60 40 141 71 155 115 170 97 47 21 38 140 154 26
125 177 19 6 197 29 132 103 178 65 145 28 149 13 146 37 116 147 42 121

Weighted regret - cycle

5 197 29 132 6 19 177 125 26 154 140 47 38 21 30 183 191 130 63 121 198 96 48 43 156 32 89
57 79 185 104 54 196 114 10 90 14 2 74 20 153 129 127 80 192 162 76 135 27 11 66 67 24 34
152 181 126 88 179 87 82 167 51 85 59 73 98 151 122 173 3 184 86 46 72 117 8 120 133 136 172
160 69 124 4 64 33 40 141 60 157 45 134 175 71 155 97 170 115 180 5

Conclusions

Regret changes the selection priority, but its effect depends on the insertion representation.

Main findings

On TSPA, weighted regret on the closed cycle obtained the best individual result (72011) and the best average among the new methods (73335.27).

On TSPB, weighted regret on the open path was strongest: minimum 51301 and average 57804.59.

Pure 2-regret performs worse on these instances because it prioritizes nodes with a large difference between their first and second insertion choices, even when their best insertion is expensive. The weighted criterion restores pressure toward a low immediate objective change.

The open-path and closed-cycle variants behave differently. The cycle variant accounts for every insertion edge in the current closed route, while the path variant follows the extended nearest-neighbor construction used in lab 01.

All 3,200 generated routes passed the internal structural validation: exactly 100 distinct selected nodes, valid indices, and a repeated first node. Saved routes and plots are available in evaluation/.

NEXT STEP

Local search can refine the constructive routes in the next laboratory.